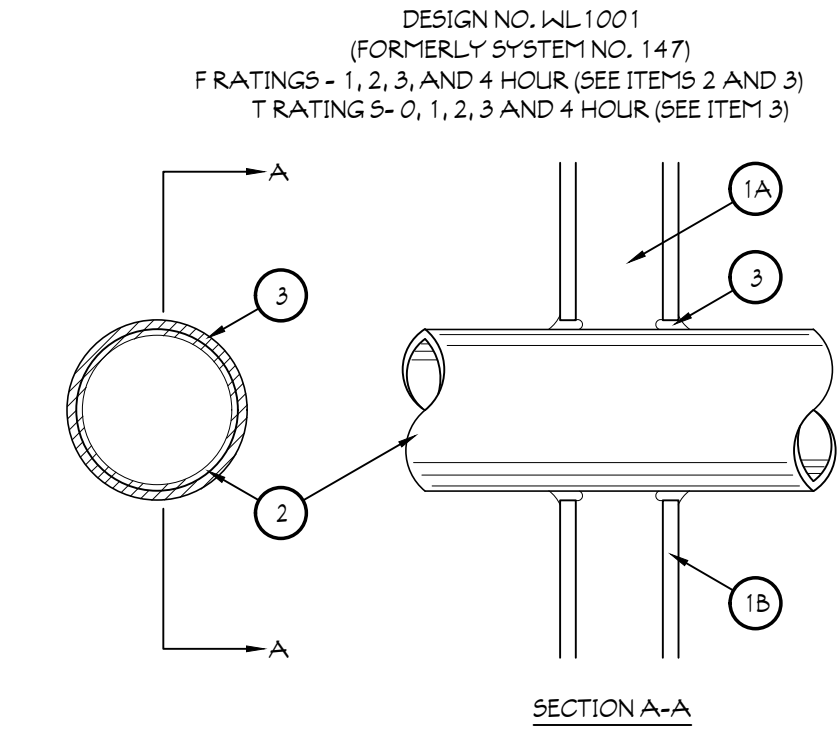


PIPE PENETRATION AT WALL



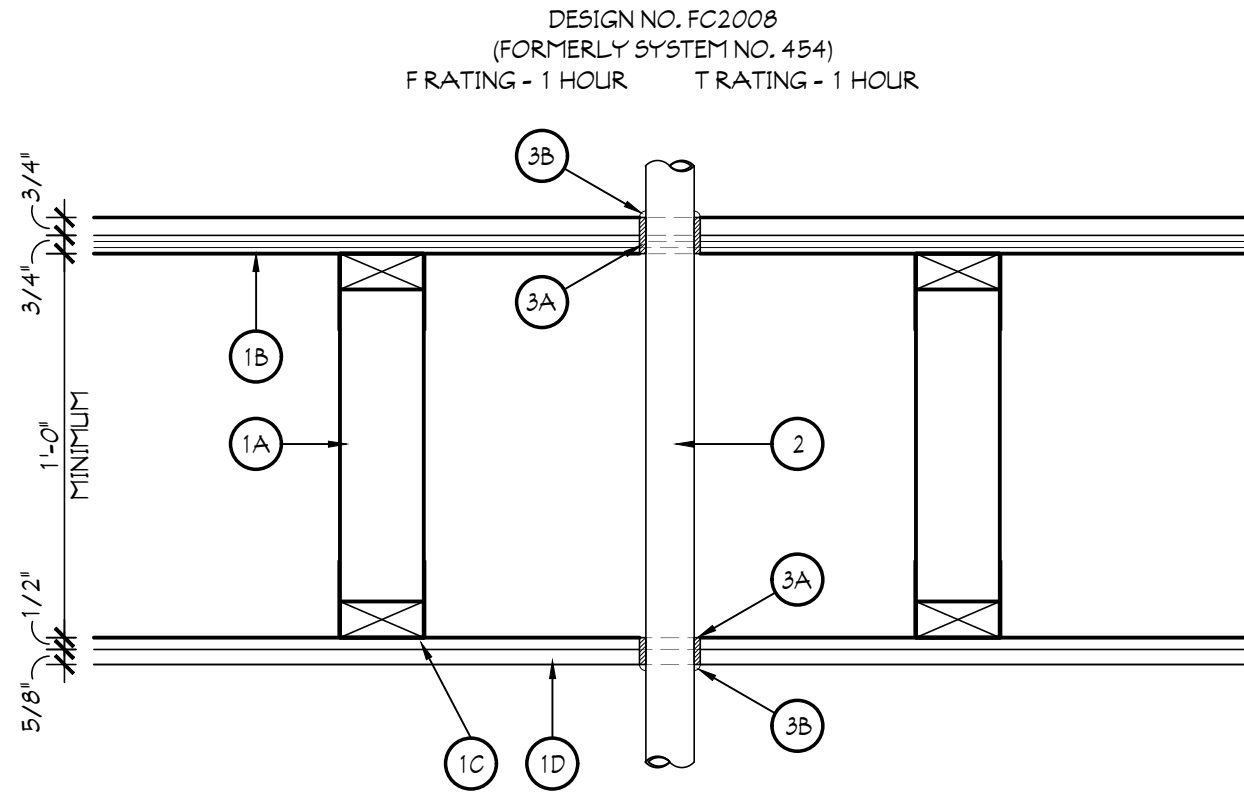
- WALL ASSEMBLY - THE 1, 2, 3, OR 4 HOUR FIRE-RATED GYPSUM WALLBOARD / STUD ASSEMBLY SHALL BE CONSTRUCTED OF THE MATERIALS AND IN THE MANNER DESCRIBED IN THE INDIVIDUAL U300 OR U400 SERIES WALL OR PARTITION DESIGNS IN THE UL FIRE RESISTANCE DIRECTORY AND SHALL INCLUDE THE FOLLOWING CONSTRUCTION FEATURES:
 - STUDS - WALL FRAMING MAY CONSIST OF EITHER WOOD STUDS (MAXIMUM 2 HOUR FIRE RATED ASSEMBLIES) OR STEEL CHANNEL STUDS. WOOD STUDS TO CONSIST OF NOMINAL 2x4 LUMBER SPACED 16 INCHES O.C. WITH NOMINAL 2x4 INCH LUMBER END PLATES AND CROSS BRACES. STEEL STUDS TO BE MINIMUM OF 3 5/8 INCH WIDE BY 1 3/8 INCH DEEP CHANNELS SPACED A MAXIMUM OF 24 INCHES O.C.
 - WALLBOARD - GYPSUM* - NOMINAL 1/2 OR 5/8 INCHES THICK, 4 FOOT WIDE WITH SQUARE OR TAPERED EDGES. THE GYPSUM WALLBOARD TYPE, THICKNESS, NUMBER OF LAYERS, FASTENER TYPE AND SHEET ORIENTATION SHALL BE AS SPECIFIED IN THE INDIVIDUAL U300 OR U400 SERIES DESIGN IN THE UL FIRE RESISTANCE DIRECTORY. MAXIMUM DIAMETER OF OPENING IS 1 3/2 INCHES.
- PIPE OR CONDUIT - NOMINAL 12 INCH DIAMETER (OR SMALLER) SCHEDULE 10 (OR HEAVIER) STEEL PIPE, NOMINAL 6 INCH DIAMETER (OR SMALLER) STEEL CONDUIT, NOMINAL 4 INCH DIAMETER (OR SMALLER) ELECTRICAL METALLIC TUBING OR TYPE L (OR HEAVIER) COPPER TUBING OR NOMINAL 1 INCH DIAMETER OR SMALLER FLEXIBLE STEEL CONDUIT. WHEN COPPER PIPE OR FLEXIBLE STEEL CONDUIT IS USED, MAXIMUM FRATING OF FIRE STOP SYSTEM (ITEM 3) IS 2 HOUR. STEEL PIPES OR CONDUITS LARGER THAN NOMINAL 4 INCH DIAMETER MAY ONLY BE USED IN WALLS CONSTRUCTED USING STEEL CHANNEL STUDS. A MAXIMUM OF ONE PIPE OR CONDUIT IS PERMITTED IN THE FIRE STOP SYSTEM. PIPE OR CONDUIT TO BE INSTALLED NEAR CENTER OF STUD CAVITY WIDTH AND TO BE RIGIDLY SUPPORTED ON BOTH SIDES OF WALL ASSEMBLY.
- FILL VOID OR CAVITY MATERIAL* - CAULK - CAULK/FILL MATERIAL INSTALLED TO COMPLETELY FILL ANNULAR SPACE BETWEEN PIPE OR CONDUIT AND GYPSUM WALLBOARD AND WITH A MINIMUM 1/4 INCH DIAMETER BEAD OF CAULK APPLIED TO PERIMETER OF PIPE OR CONDUIT AT ITS EGRESS FROM THE WALL. CAULK INSTALLED SYMMETRICALLY ON BOTH SIDES OF WALL ASSEMBLY. THE HOURLY FIRE RATING OF THE FIRE STOP SYSTEM IS DEPENDENT UPON THE HOURLY FIRE RATING OF THE WALL ASSEMBLY IN WHICH IT IS INSTALLED, AS SHOWN IN THE FOLLOWING TABLE. THE HOURLY T RATING OF THE FIRE STOP SYSTEM IS DEPENDENT UPON THE TYPE OR SIZE OF THE PIPE OR CONDUIT AND THE HOURLY FIRE RATING OF THE WALL ASSEMBLY IN WHICH IT IS INSTALLED, AS TABULATED BELOW:

MAXIMUM PIPE OR CONDUIT DIAMETER INCHES	ANNULAR SPACE INCHES	FRATING HOUR	T RATING HOUR
1	0 TO 3/16	1 OR 2	0+, 1 OR 2
1	1/4 TO 1/2	3 OR 4	3 OR 4
4	0 TO 1/4	1 OR 2	0
6	1/4 TO 1/2	3 OR 4	0
12	3/16 TO 3/8	1 OR 2	0

+ WHEN COPPER PIPE IS USED, T RATING IS 0 HOURS.
MINNESOTA MINING MANUFACTURING COMPANY - TYPES CP-23 S/L, CP-25 W/B, CP-25 W/B+

*BEARING THE UL CLASSIFICATION MARKING

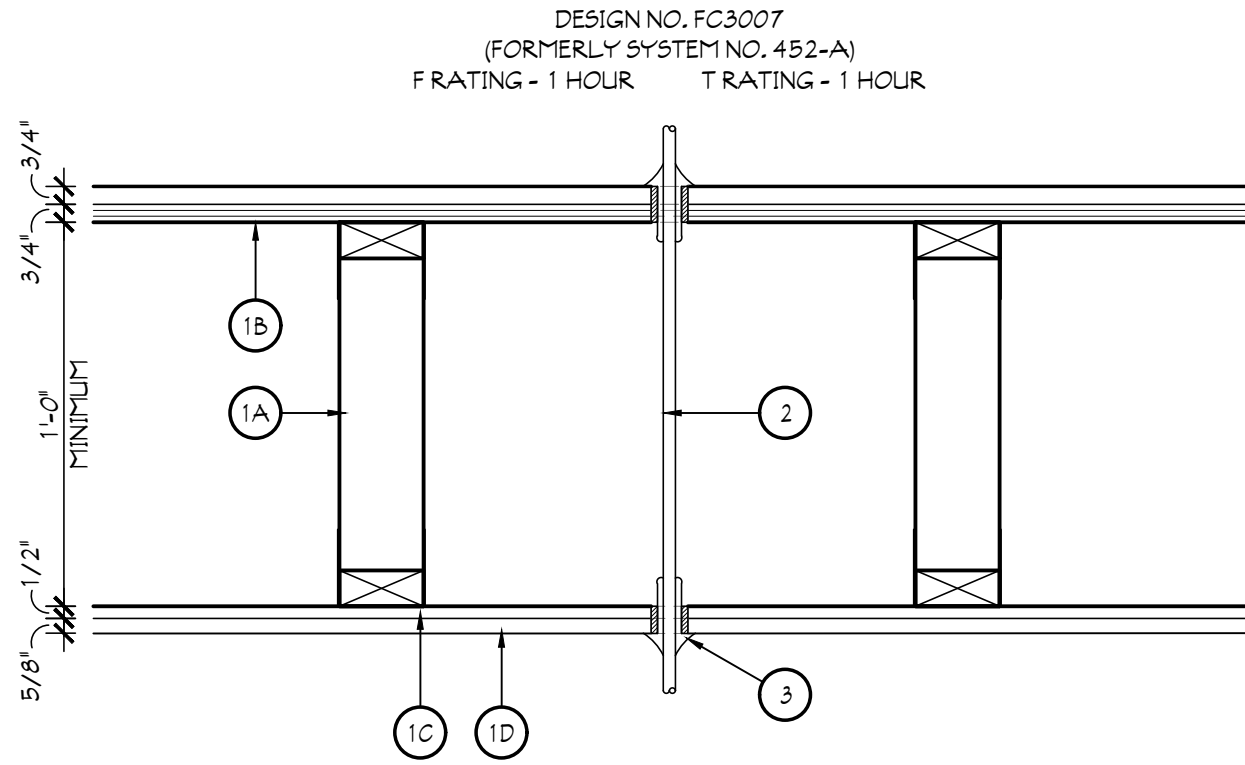
PIPE PENETRATION AT CEILING / FLOOR



- FLOOR ASSEMBLY - THE FIRE RATED WOOD TRUSS OR COMBINATION WOOD AND STEEL TRUSS FLOOR - CEILING ASSEMBLY SHALL BE CONSTRUCTED OF THE MATERIALS AND IN THE MANNER DESCRIBED IN THE INDIVIDUAL L500 SERIES DESIGN IN THE UL FIRE RESISTANCE DIRECTORY AND SHALL INCLUDE THE FOLLOWING CONSTRUCTION FEATURES:
 - TRUSSES - MINIMUM 12" DEEP PARALLEL CHORD TRUSSES FABRICATED FROM NOMINAL 2x4 INCH LUMBER IN CONJUNCTION WITH GALVANIZED STEEL TRUSS PLATES OR STRUCTURAL WOOD MEMBERS* WITH BRIDGING AS REQUIRED.
 - FLOORING - NOMINAL 3/4 INCH THICK PLYWOOD FLOORING WITH OR WITHOUT FLOOR TOPPING MIXTURE*, MAXIMUM DIAMETER OF OPENING HOLE-SAWED IN FLOORING IS 3 INCHES.
 - FURRING CHANNELS - RIGID OR RESILIENT GALVANIZED STEEL FURRING CHANNELS INSTALLED PERPENDICULAR TO BOTTOM CHORD OF TRUSSES.
 - WALL BOARD, GYPSUM* - NOMINAL 4 FEET WIDE BY 5/8 INCH THICK, SCREW ATTACHED TO FURRING CHANNELS. MAXIMUM DIAMETER OF HOLE-SAWED OPENING IN GYPSUM WALLBOARD CEILING IS 3 INCHES.
- NONMETALLIC PIPE - NOMINAL 2 INCH DIAMETER (OR SMALLER) SCHEDULE 40 POLY-VINYL CHLORIDE (PVC), SDR 17 CHLORINATED POLY-VINYL CHLORIDE (CPVC) OR SOLID-GORE SCHEDULE 40 ACRYLONITRILE - BUTADIENE - STYRENE (ABS) PIPE FOR USE IN CLOSED (PROCESS OR SUPPLY) OR VENTED (DRAIN, WASTE, OR VENT) PIPING SYSTEMS. ONE PIPE TO BE INSTALLED APPROXIMATELY MID-WAY BETWEEN TRUSSES AND CENTERED IN CIRCULAR OPENINGS IN FLOORING AND IN CEILING. A NOMINAL 1/4 TO 3/8 INCH ANNULAR SPACE IS REQUIRED IN THE FIRE STOP SYSTEM. PIPE TO BE RIGIDLY SUPPORTED ON BOTH SIDES OF FLOOR - CEILING ASSEMBLY.
- FIRE STOP SYSTEM - THE DETAILS OF THE FIRE STOP SYSTEM SHALL BE AS FOLLOWS:
 - FILL VOID OR CAVITY MATERIALS* - WRAP STRIP - NOMINAL 1/4 INCH THICK INTUMESCENT ELASTOMERIC MATERIAL FACED ONE SIDE WITH ALUMINUM FOIL, SUPPLIED TO 2 INCH WIDE STRIPS. NOMINAL 2 INCH WIDE STRIP TIGHTLY WRAPPED AROUND NONMETALLIC PIPE (FOIL SIDE EXPOSED), SECURED WITH TWO STEEL TIE WIRES AND SLID TO HOLE-SAWED OPENING IN FLOORING (ITEM 1B) AND IN GYPSUM WALLBOARD CEILING (ITEM 1D). BOTTOM EDGE OF WRAP STRIP TO PROJECT 4/16 TO 11/16 INCHES BELOW BOTTOM SURFACE OF FLOORING AND BELOW BOTTOM CEILING SURFACE OF GYPSUM WALLBOARD.
MINNESOTA MINING AND MANUFACTURING COMPANY - TYPE PS-145
 - FILL VOID OR CAVITY MATERIALS* - CAULK OR PUTTY NOMINAL 1/4 INCH THICKNESS OF CAULK OR PUTTY TO BE APPLIED TO THE EXPOSED EDGES OF THE WRAP STRIP LAYER (TOP OF FLOORING AND BOTTOM OF GYPSUM WALLBOARD CEILING). GENEROUS APPLICATION OF CAULK OR PUTTY TO BE APPLIED TO FILL ALL GAPS AT THE WRAP STRIP / FLOORING AND WRAP STRIP / GYPSUM WALL BOARD CEILING INTERFACES.
MINNESOTA MINING AND MANUFACTURING COMPANY - TYPE CP-25 W/B, CP-25 W/B+, TYPE MP PUTTY

*BEARING THE UL CLASSIFICATION MARKING

CABLE PENETRATION AT CEILING / FLOOR



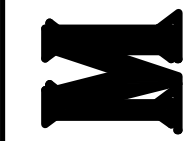
- FLOOR ASSEMBLY - THE FIRE RATED WOOD TRUSS OR COMBINATION WOOD AND STEEL TRUSS FLOOR - CEILING ASSEMBLY SHALL BE CONSTRUCTED OF THE MATERIALS AND IN THE MANNER DESCRIBED IN THE INDIVIDUAL L500 SERIES DESIGN IN THE UL FIRE RESISTANCE DIRECTORY AND SHALL INCLUDE THE FOLLOWING CONSTRUCTION FEATURES:
 - TRUSSES - MINIMUM 12" DEEP PARALLEL CHORD TRUSSES FABRICATED FROM NOMINAL 2x4 INCH LUMBER IN CONJUNCTION WITH GALVANIZED STEEL TRUSS PLATES OR STRUCTURAL WOOD MEMBERS* WITH BRIDGING AS REQUIRED.
 - FLOORING - NOMINAL 3/4 INCH THICK PLYWOOD FLOORING WITH OR WITHOUT FLOOR TOPPING MIXTURE*, MAXIMUM DIAMETER OF OPENING HOLE-SAWED IN FLOORING IS 1 INCH.
 - FURRING CHANNELS - RIGID OR RESILIENT GALVANIZED STEEL FURRING CHANNELS INSTALLED PERPENDICULAR TO BOTTOM CHORD OF TRUSSES.
 - WALL BOARD, GYPSUM* - NOMINAL 4 FEET WIDE BY 5/8 INCH THICK, SCREW ATTACHED TO FURRING CHANNELS. MAXIMUM DIAMETER OF HOLE-SAWED OPENING IN GYPSUM WALLBOARD CEILING IS 1 INCHES.
- CABLES - MAXIMUM SEVEN CONDUCTOR NO. 12 AWG (OR SMALLER) POWER / CONTROL CABLES OR MAXIMUM 100 PAIR NO. 22 AWG (OR SMALLER) TELECOMMUNICATION CABLES WITH POLY-VINYL CHLORIDE INSULATION AND JACKET MATERIALS. INDIVIDUAL CABLES TO BE CENTERED IN NOMINAL 1 INCH DIAMETER CIRCULAR CUT OUTS. CABLES TO BE RIGIDLY SUPPORTED ON BOTH SIDES OF FLOOR - CEILING ASSEMBLY.
- FILL VOID OR CAVITY MATERIALS* - CAULK OR PUTTY - CAULK OR PUTTY FILL MATERIAL FORCED INTO ANNULAR SPACE THROUGHOUT THICKNESS OF FLOORING AND GYPSUM WALLBOARD CEILING AND WITH A MINIMUM 1/4 INCH BEAD APPLIED TO PERIMETER OF CABLE AT ITS EGRESS FROM THE TOP OF THE FLOORING AND THE UNDERSIDE OF THE GYPSUM WALLBOARD CEILING.
MINNESOTA MINING AND MANUFACTURING COMPANY - TYPE CP-25 W/B

*BEARING THE UL CLASSIFICATION MARKING

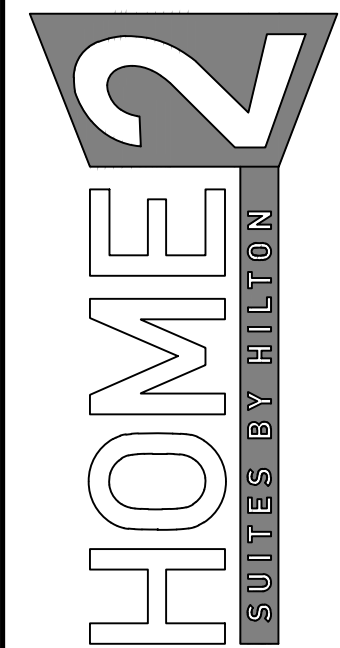
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ALL DIMENSIONS UNLESS OTHERWISE SPECIFIED SHALL BE IN INCHES. DIMENSIONS SHALL NOT BE HELD RESPONSIBLE FOR ANY GENERAL CONTRACTOR. MUST ASSESS THE CONSTRUCTION OF THE PROJECT PRIOR TO CONSTRUCTION.

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SHEET SIZE 24x36

FT. WAYNE, INDIANA

REVISIONS:

DATE: 05/01/26

JOB NO.: MA9-26-105

NAME:

FLOOR AND PIPE

PENETRATIONS DETAILS

SHEET:

A454